Dear Math 142 Students,

- This package contains solution keys for your assignments 1 — 6.
- It also contains some solutions to other interesting problems.
- I hope it will be useful for your review for the Midterm I.

With my Care,

W. Gu

Chapter 1 Curves

Prof. Gu

1-2 Parametrized Curve

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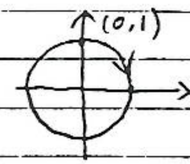
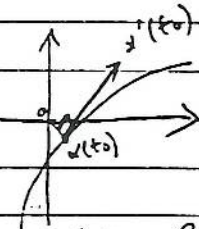
#1 Let $\alpha(t) = (\sin t, \cos t)$, then



$x^2(t) + y^2(t) = \sin^2 t + \cos^2 t = 1$

$\alpha(0) = (0, 1)$ Q.E.D

#2 Let $l(t) = \|\alpha(t)\|^2$



To minimize $l(t)$ is equivalent to minimize $\|\alpha(t)\|$.

Spse $l(t)$ achieves its minimum at t_0, then $l'(t_0)$ must equal to zero.

Now $l(t) = \alpha(t) \cdot \alpha(t)$

$l'(t) = \alpha'(t) \cdot \alpha(t) + \alpha(t) \cdot \alpha'(t) = 2\alpha'(t) \cdot \alpha(t)$

$\Rightarrow l'(t_0) = 2\alpha'(t_0) \cdot \alpha(t_0) = 0$

$\Rightarrow \alpha'(t_0) \cdot \alpha(t_0) = 0$

$\Rightarrow \alpha'(t_0) \perp \alpha(t_0)$

Q.E.D

#3 Suppose $t \in (a, b)$, then $\alpha(t)$ is a line segment where $t \in (a, b)$.

Pf: $\alpha''(t) = 0$

$\Rightarrow \alpha'(t) = \vec{a}$ (constant vector!)

$\Rightarrow \alpha(t) = \vec{a}t + \vec{b}$, b (constant vector!). Q.E.D

#4 Let $\alpha(t) = (x(t), y(t), z(t))$, $t \in I$, $0 \in I$

$\alpha'(t) = (x'(t), y'(t), z'(t))$, $V = (a, b, c)$

Now $\alpha'(t) \cdot V = 0$ $\qquad \alpha(t) \perp V$

$\Rightarrow a x'(t) + b y'(t) + c z'(t) = 0$ $\qquad \Uparrow$

$\Rightarrow a\int_0^t x'(t)\,dt + b\int_0^t y'(t) + c\int_0^t z'(t)\,dt = 0$ $\qquad \alpha(t) \cdot V = 0$

$\Rightarrow a\, x(t)\Big|_0^t + b\, y(t)\Big|_0^t + c\, z(t)\Big|_0^t = 0$ $\qquad \Uparrow$

$\Rightarrow a x(t) + b y(t) + c z(t) - a x(0) - b y(0) - c z(0) = 0 \Rightarrow \alpha(t) \cdot V - \underbrace{\alpha(0) \cdot V}_{0} = 0$

#5 Spse $|\alpha(t)|$ is nonzero constant.

$\Rightarrow |\alpha(t)|^2 = c$ (constant)

$\Rightarrow \alpha(t) \cdot \alpha(t) = 0$

$\Rightarrow \alpha'(t) \cdot \alpha(t) + \alpha(t) \cdot \alpha'(t) = 0$

$\Rightarrow 2\alpha'(t) \cdot \alpha(t) = 0$

$\Rightarrow \alpha'(t) \perp \alpha(t)$

Conversely, spse $\alpha(t) \perp \alpha'(t)$. Let $\alpha(t) = (x(t), y(t), z(t))$

Then $x(t)\,x'(t) + y(t)\,y'(t) + z(t)\,z'(t) = 0$

$$\Rightarrow \int_{t_0}^{t} x(t)\,x'(t)\,dt + \int_{t_0}^{t} y(t)\,y'(t)\,dt + \int_{t_0}^{t} z(t)\,z'(t)\,dt = 0$$

$$\Rightarrow \int_{t_0}^{t} x(t)\,dx(t)\,dt + \int_{t_0}^{t} y(t)\,dy(t) + \int_{t_0}^{t} z(t)\,dz(t) = 0$$

$$\Rightarrow \frac{x^2(t)}{2} + \frac{y^2(t)}{2} + \frac{z^2(t)}{2} = \frac{x^2(t_0)}{2} + \frac{y^2(t_0)}{2} + \frac{z^2(t_0)}{2}$$

Let $A = x^2(t_0) + y^2(t_0) + z^2(t_0)$ (a constant)

$\cancel{\Rightarrow x^2(t) + y^2(t) + z^2(t) = A}$

$\Rightarrow |\alpha(t)|^2 = A \Rightarrow |\alpha(t)| = A$ Q.E.D

Prof. Gu

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Section 1-3. Regular Curves ; Arc Length

2. A circular disk of radius 1 in the plane xy rolls without slipping along the x axis. The figure described by a point of the circumference of the disk is called a *cycloid* (Fig. 1-7).

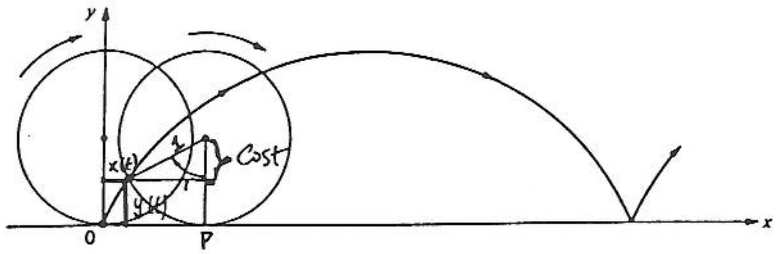

Figure 1-7. The cycloid.

*a. Obtain a parametrized curve $\alpha: R \to R^2$ the trace of which is the cycloid, and determine its singular points.

b. Compute the arc length of the cycloid corresponding to a complete rotation of the disk.

a) Let's first caculate op.
Since $l = rt = 1\cdot t = t$ and the disk rolls without slipping
so we have $op = l = t$
$\Rightarrow x(t) = t - \sin t$
clearly $y(t) = 1 - \cos t$
Thus $\alpha(t) = (t - \sin t,\ 1 - \cos t)$. Singular points : $t = 2\pi k,\ k \in \mathbb{Z}$

b) $\int_0^{2\pi} |\alpha'(t)|\, dt = \int_0^{2\pi} \sqrt{(1-\cos t)^2 + (\sin t)^2}\; dt$

$= \int_0^{2\pi} \sqrt{2(1-\cos t)}\, dt = \int_0^{2\pi} \sqrt{4 \sin^2 \frac{t}{2}}\, dt \;\overset{\sin\frac{t}{2} \geq 0 \text{ over } [0,2\pi]}{=\!=\!=}\; \int_0^{2\pi} 2 \sin \frac{t}{2}\, dt$

$= -4 \cos \frac{t}{2} \Big|_0^{2\pi} = -(0-4) = 4$

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3. Let $0A = 2a$ be the diameter of a circle S^1 and $0Y$ and AV be the tangents to S^1 at 0 and A, respectively. A half-line r is drawn from 0 which meets the circle S^1 at C and the line AV at B. On $0B$ mark off the segment $0p = CB$. If we rotate r about 0, the point p will describe a curve called the *cissoid of Diocles*. By taking $0A$ as the x axis and $0Y$ as the y axis, prove that

a. The trace of

$$\alpha(t) = \left(\frac{2at^2}{1+t^2}, \frac{2at^3}{1+t^2}\right), \qquad t \in R,$$

is the cissoid of Diocles ($t = \tan\theta$; see Fig. 1-8).

b. The origin (0, 0) is a singular point of the cissoid.

c. As $t \to \infty$, $\alpha(t)$ approaches the line $x = 2a$, and $\alpha'(t) \to (0, 2a)$. Thus, as $t \to \infty$, the curve and its tangent approach the line $x = 2a$; we say that $x = 2a$ is an *asymptote* to the cissoid.

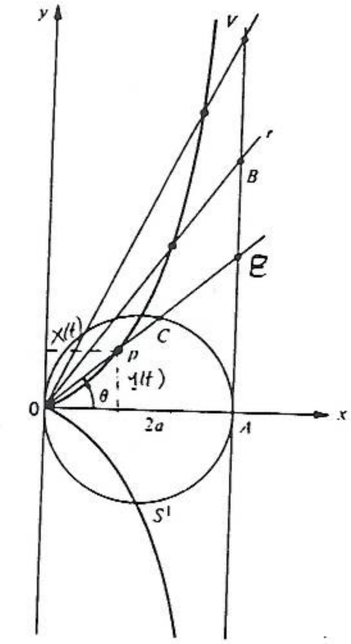

Figure 1-8. The cissoid of Diocles.

a) Let's first find the coordinates for B:

$$\frac{y}{2a} = \tan\theta \Rightarrow y = 2a\tan\theta$$

and x is clearly $2a$

$\Rightarrow B: (2a, 2a\tan\theta)$

Next, let's compute the coordinate for C:

Since the circle has equation $(x-a)^2+y^2=a^2$, the line pass OB is $y=(\tan\theta)x$, thus their intersection is a solution to

$$(x-a)^2 + ((\tan\theta)x)^2 = a^2$$

$$\Rightarrow x^2 - 2ax + \cancel{a^2} + (\tan\theta)^2x^2 = \cancel{a^2}$$

$$\Rightarrow x^2(1+\tan^2\theta) - 2ax = 0$$

$$\Rightarrow x^2\sec^2\theta - 2ax = 0$$

$$\Rightarrow x(x\sec^2\theta - 2a) = 0 \Rightarrow x = 0 \text{ or } x = \frac{2a}{\sec^2\theta} = 2a\cos^2$$

Thus $C: (2a\cos^2\theta \quad 2a\cos^2\theta\tan\theta)$

Now $\overrightarrow{OP} = \overrightarrow{OB} - \overrightarrow{OC} = (2a - 2a\cos^2\theta,\ 2a\tan\theta - 2a\cos^2\theta\tan\theta)$

$= (2a(\sin^2\theta) \quad 2a\tan\theta\sin^2\theta)$ (*)

Let $t = \tan\theta$

Now $\dfrac{t^2}{1+t^2} = \dfrac{\tan^2\theta}{1+\tan^2\theta} = \dfrac{\tan^2\theta}{\sec^2\theta} = \dfrac{\frac{\sin^2\theta}{\cancel{\cos^2\theta}}}{\frac{1}{\cancel{\cos^2\theta}}} = \sin^2\theta$

Thus $\vec{OP} = \left(\dfrac{2at^2}{1+t^2}, \dfrac{2at^3}{1+t^2}\right)$

Another way to show a) is by using polor coordinates:

Since the circle is $(x-a)^2 + y^2 = a^2 \Longrightarrow \begin{array}{l} x = a\cos\varphi + a \\ y = a\sin\varphi \end{array}$

Note when φ run from 0 to π, θ is from 0 to $\frac{\pi}{2}$

Thus here $\varphi = 2\theta$.

Now the length of OC is $r_C = \sqrt{(a\cos2\theta + a)^2 + (a\sin2\theta)^2}$

$= a^2\sqrt{\cos^2 2\theta + 2\cos2\theta + 1 + (\sin 2\theta)^2}$

$= a^2\sqrt{2(1+\cos2\theta)}$

$= a^2\sqrt{4\cos^2\theta} = 2a\cos\theta$

Now since the line is $x = 2a \Rightarrow r\cos\theta = 2a \Rightarrow r = 2a\sec\theta$

The length of OB is $r_B = 2a\sec\theta$

Now the length of OP is $r_p = r_B - r_C = 2a\sec\theta - 2a\cos\theta$

Now, the coordinates of P is $(r_p\cos\theta, r_p\sin\theta)$

$= \left(\left(2a\frac{1}{\cos\theta} - 2a\cos\theta\right)\cos\theta, \left(2a\frac{1}{\cos\theta} - 2a\cos\theta\right)\sin\theta\right)$

$= \left(2a(1-\cos^2\theta), 2a\tan\theta - 2a\tan\theta\cos^2\theta\right)$

$= (2a\sin^2\theta, 2a\tan\theta\sin^2\theta)$ same as (*)

3b) $\alpha(t) = \left(\frac{2at^2}{1+t^2}, \frac{2at^3}{1+t^2} \right)$

$$\alpha'(t) = \left(\frac{4at(1+t^2) - 2t(2at^2)}{(1+t^2)^2}, \frac{6at^2(1+t^2) - 2t(2at^3)}{(1+t^2)^2} \right)$$

$$= \left(\frac{4at}{(1+t^2)^2}, \frac{6at^2+2at^4}{(1+t^2)^2} \right)$$

$$= \left(\frac{4at}{(1+t^2)^2}, \frac{2at^2(3+t^2)}{(1+t^2)^2} \right)$$

$$\begin{cases} 4at = 0 \\ 2at^2(3+t^2) = 0 \end{cases} \overset{a \neq 0}{\Rightarrow} t = 0 \quad \Rightarrow \alpha(0) = (0, 0)$$

Thus the origin $(0, 0)$ is a singular pt of the cissoid.

3c) as $t \to \infty$ $\quad y(t) = \frac{2at^3}{1+t^2} \xrightarrow{t \to \infty} \infty$

$$x(t) = \frac{2at^2}{1+t^2} = \frac{2a(1+t^2) - 2a}{1+t^2} = 2a - \frac{2a}{1+t^2} \xrightarrow{t \to \infty} 2a$$

Thus as $t \to \infty$, $\alpha(t)$ approaches to the line $x = 2a$.

By 3b), $\alpha'(t) = \left(\frac{4at}{(1+t^2)^2}, \frac{6at^2+2at^4}{(1+t^2)^2} \right) \longrightarrow (0, 2a)$,

i.e $\frac{y(t)}{x(t)} \xrightarrow{t \to \infty} \infty$

Thus, as $t \to \infty$, the curve and its tangent approach the line $x = 2a$.

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(5.) Let $\alpha: (-1, +\infty) \longrightarrow R^2$ be given by

$$\alpha(t) = \left(\frac{3at}{1+t^3}, \frac{3at^2}{1+t^3}\right).$$

Prove that:

a. For $t = 0$, α is tangent to the x axis.

b. As $t \longrightarrow +\infty$, $\alpha(t) \longrightarrow (0, 0)$ and $\alpha'(t) \longrightarrow (0, 0)$.

c. Take the curve with the opposite orientation. Now, as $t \longrightarrow -1$, the curve and its tangent approach the line $x + y + a = 0$.

The figure obtained by completing the trace of α in such a way that it becomes symmetric relative to the line $y = x$ is called the *folium of Descartes* (see Fig. 1-10).

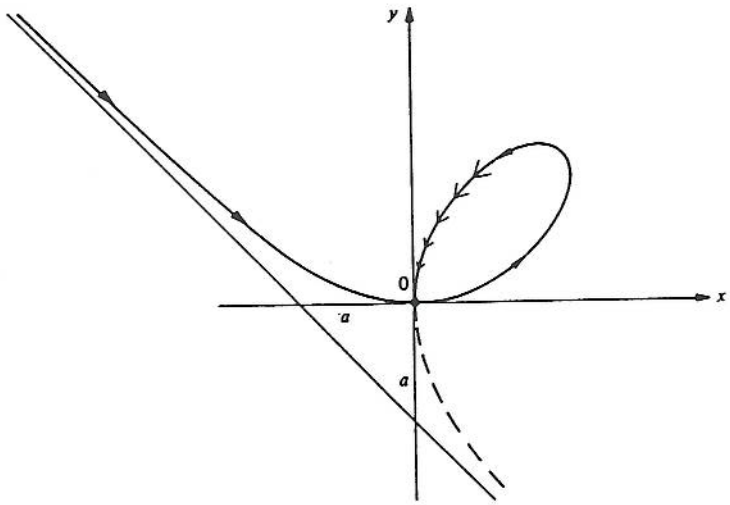
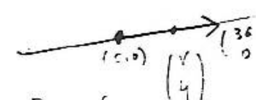


Figure 1-10. Folium of Descartes.

a) $$\alpha'(t) = \left(\frac{3a(1+t^3) - 3t^2(3at)}{(1+t^3)^2}, \frac{6at(1+t^3) - 3t^2(3at^2)}{(1+t^3)^2}\right)$$

$$= \left(\frac{3a - 6at^3}{(1+t^3)^2}, \frac{6at - 3at^4}{(1+t^3)^2}\right) \xrightarrow{t \to 0} (3a, 0)$$



Note $\alpha(0,0) = (0, 0)$.

So the tangent line is $\begin{pmatrix} x-0 \\ y-0 \end{pmatrix} = t\begin{pmatrix} 3a \\ 0 \end{pmatrix} \Rightarrow \begin{cases} x = 3at \\ y = 0 \end{cases}$

$\Rightarrow$ For $t=0$, α is tangent to the x axis.

b) It is clearly that $\alpha(t) \xrightarrow{t \to \infty} (0, 0)$

By a), we see $\alpha'(t) \xrightarrow{t \to \infty} (0, 0)$

c) $$x(t) + y(t) + a = \frac{3at}{1+t^3} + \frac{3at^2}{1+t^3} + a = \frac{3at + 3at^2 + a(1+t^3)}{1+t^3}$$

$$= \frac{a(1+t)^3}{1+t^3} \xrightarrow{t \to -1} \frac{3a(1+t)^2}{3t^2} \xrightarrow{t \to -1} 0.$$

Thus the curve tend to the line $x + y + a = 0$

$\frac{0}{0}$ type

and $\lim_{t\to -1} \frac{y(t)}{x(t)} = \lim_{t\to -1} \frac{\frac{3at^2}{\cancel{1+t^3}}}{\frac{3at}{\cancel{1+t^3}}} = \lim_{t\to -1} t = -1$

Thus the slope of the tangent line is -1

Since α approaches the line $x+y+a=0$ with slope also -1

Thus α and its tangent both approach to $x+y+a=0$

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6. Let $\alpha(t) = (ae^{bt}\cos t, ae^{bt}\sin t)$, $t \in R$, a and b constants, $a > 0$, $b < 0$, be a parametrized curve.

a. Show that as $t \to +\infty$, $\alpha(t)$ approaches the origin 0, spiraling around it (because of this, the trace of α is called the *logarithmic spiral*; see Fig. 1-11).

b. Show that $\alpha'(t) \to (0, 0)$ as $t \to +\infty$ and that

$$\lim_{t\to+\infty} \int_{t_0}^{t} |\alpha'(t)|\, dt$$

is finite; that is, α has finite arc length in $[t_0, \infty)$.

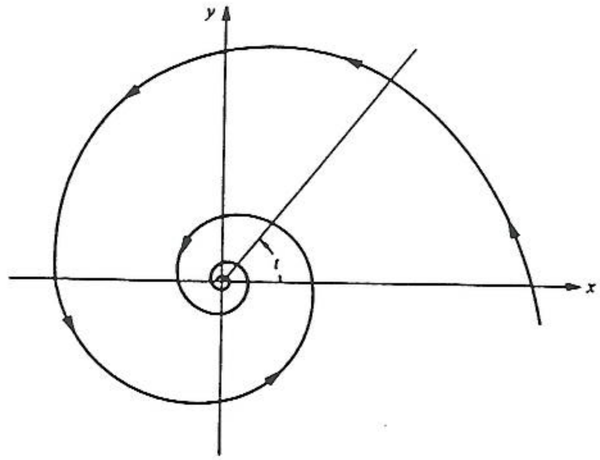


Figure 1-11. Logarithmic spiral.

a) $\alpha(t) = (ae^{bt}\cos t, ae^{bt}\sin t)$

$$= \left(\frac{a\cos t}{e^{-bt}}, \frac{a\sin t}{e^{-bt}}\right) \xrightarrow{t\to\infty} (0, 0)$$

since $|a\sin t| \leq a$, $|a\sin t| \leq a$ $(a>0)$

and $\frac{1}{e^{-bt}} \xrightarrow[b<0]{t\to+\infty} 0$

b) $\alpha'(t) = (abe^{bt}\cos t - ae^{bt}\sin t, abe^{bt}\sin t + ae^{bt}\cos t)$

$$= \left(e^{bt}(ab\cos t - a\sin t), e^{bt}(ab\sin t + a\cos t)\right)$$

$\xrightarrow{t\to+\infty} (0, 0)$ since $|ab\cos t - a\sin t| \leq |ab||\cos t| + |a||\sin t|$

$\leq a|b| + a$, $a>0$

$|ab\sin t + a\cos t| \leq a|b| + a$, $a>0$

Let's calculate

$$\lim_{t\to+\infty}\int_{t_0}^{t} |\alpha'(t)|\,dt.$$

$$|\alpha'(t)|^2 = a^2(e^{bt})^2\left[(b\cos t-\sin t)^2+(b\sin t+\cos t)^2\right]$$

$$= a^2(e^{bt})^2\left(\begin{array}{l} b^2\cos^2 t - \cancel{2b\sin t\cos t}+\sin^2 t \\ +b^2\sin^2 t+\cancel{2b\sin t\cos t}+\cos^2 t\end{array}\right)$$

$$= a^2(e^{bt})^2\left[b^2+1\right]$$

$$|\alpha'(t)| = a\sqrt{b^2+1}\,e^{bt}$$

Now $\displaystyle\lim_{t\to+\infty}\int_{t_0}^{t} a\sqrt{b^2+1}\,e^{bt}dt = \lim_{\substack{t\to+\infty \\ (b<0)}}\frac{a}{b}\sqrt{b^2+1}\,e^{bt}\Big|_{t_0}^{t}$

$$= \lim_{t\to+\infty}\left(\frac{a}{b}\sqrt{b^2+1}\,e^{bt}-\frac{a}{b}\sqrt{b^2+1}\,e^{bt_0}\right)$$

$$= \frac{a}{(-b)}\sqrt{b^2+1}\,e^{bt_0}$$

Thus α has finite length in $[t_0,+\infty)$.

From Lecture:

Exercise: The viviani's curve is defined to be intersection of the sphere $x^2+y^2+z^2=4$ and the cylinder $(x-1)^2+y^2=1$. Show that the length of the vivani curve is great than the length of the equator of the sphere.

Soln 1
$$\begin{cases} x^2+y^2+z^2=4 & \cdots (1) \\ (x-1)^2+y^2=1 & \cdots (2) \end{cases}$$

para the cylinder $\Rightarrow \begin{cases} x-1=\cos\theta \\ y=\sin\theta \\ z=z \end{cases}$ (*)

Please see the picture on next page!

(*) plugs into (1),

$$(1+\cos\theta)^2+\sin^2\theta+z^2=4$$

$$\Rightarrow 1+2\cos\theta+\underbrace{\cos^2\theta+\sin^2\theta}_{1}+z^2=4$$

$$\Rightarrow z^2=2-2\cos\theta=2(1-\cos\theta)=2^2\sin^2\frac{\theta}{2}$$

$$\Rightarrow z=\pm 2\sin\theta/2$$

Thus (*) $\Rightarrow \begin{cases} x=1+\cos\theta \\ y=\sin\theta \\ z=2\sin\theta/2 \end{cases}$

So $\alpha(\theta)=(1+\cos\theta,\ \sin\theta,\ 2\sin\theta/2)$ is the intersection

$$\alpha'(\theta)=(-\sin\theta,\ \cos\theta,\ \cancel{2}\tfrac{1}{\cancel{2}}\cos\theta/2)$$

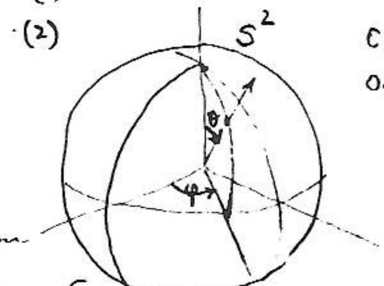
$$|\alpha'(\theta)|=\sqrt{\sin^2\theta+\cos^2\theta+\cos^2\theta/2}=\sqrt{1+\cos^2\theta/2}$$

see next page

$$\ell(\alpha)=\int_0^{2\pi}\sqrt{1+\cos^2\theta/2}\,d\theta \doteq 15.2872>4\pi=12.566$$

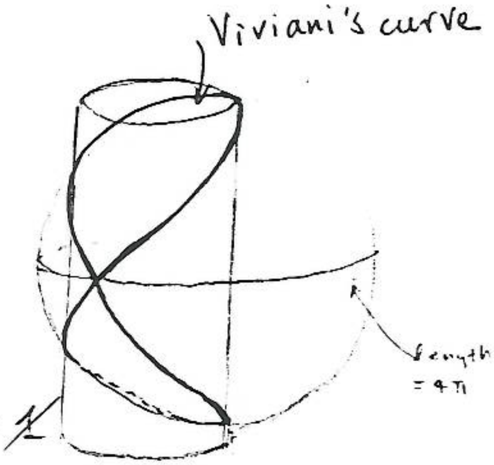
* See Solution 2 on next page!

Solun 2 $\begin{cases} x^2+y^2+z^2=4 & \cdots(1) \\ (x-1)^2+y^2=1 & \cdots(2) \end{cases}$



$0 \le \theta \le \pi$

$0 \le \varphi \le 2\pi$



let $\begin{cases} x = 2\sin\theta\cos\varphi \\ y = 2\sin\theta\sin\varphi \\ z = 2\cos\theta \end{cases}$

be the parametrization of S^2.

Plug into (2): $(2\sin\theta\cos\varphi - 1)^2 + (2\sin\theta\sin\varphi)^2 = 1$

$\Rightarrow 4\sin^2\theta\cos^2\varphi - 4\sin\theta\cos\varphi + \cancel{1} + 4\sin^2\theta\sin^2\varphi = \cancel{1}$

also $= 4\pi$

$\Rightarrow \sin^2\theta - \sin\theta\cos\varphi = 0$

$\Rightarrow \sin\theta(\sin\theta - \cos\varphi) = 0$ W.L.G. assume $0 < \theta < \pi$

$\Rightarrow \sin\theta = \cos\varphi$

$$\begin{cases} x = 2\sin^2\theta \\ y = 2\sin\theta\sqrt{1-\cos^2\varphi} = 2\sin\theta\underbrace{\sqrt{1-\sin^2\theta}}_{\cos\theta} = 2\sin\theta\cos\theta = \sin 2\theta, \quad 0 \le \theta \le \frac{\pi}{2} \\ z = 2\cos\theta \end{cases}$$

$\alpha(\theta) = (2\sin^2\theta,\ \sin 2\theta,\ 2\cos\theta)$

$\alpha'(\theta) = (\underbrace{4\sin\theta\cos\theta}_{2\sin 2\theta},\ 2\cos 2\theta,\ -2\sin\theta)$

$|\alpha'(\theta)| = \sqrt{(2\sin 2\theta)^2 + (2\cos 2\theta)^2 + (-2\sin\theta)^2}$

$= \sqrt{4 + 4\sin^2\theta}$

$\because 0 \le \theta \le \frac{\pi}{2}$

$= \sqrt{4(1+\sin^2\theta)} = 2\sqrt{1+\sin^2\theta}$

$l(\alpha) = 4\int_0^{\frac{\pi}{2}} 2\sqrt{1+\sin^2\theta}\,d\theta = 15.2872$

$\int_0^{\frac{\pi}{2}} \sqrt{1+\sin^2\theta}\,d\theta =$ elliptic $E(-1)$
$= 1.9109$

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(*8.) Let $\alpha: I \longrightarrow R^3$ be a differentiable curve and let $[a, b] \subset I$ be a closed interval. For every *partition*

$$a = t_0 < t_1 < \cdots < t_n = b$$

of $[a, b]$, consider the sum $\sum_{i=1}^{n} |\alpha(t_i) - \alpha(t_{i-1})| = l(\alpha, P)$, where P stands for the given partition. The norm $|P|$ of a partition P is defined as

$$|P| = \max(t_i - t_{i-1}),\ i = 1, \ldots, n.$$

Geometrically, $l(\alpha, P)$ is the length of a polygon inscribed in $\alpha([a, b])$ with vertices in $\alpha(t_i)$ (see Fig. 1-12). The point of the exercise is to show that the arc length of $\alpha([a, b])$ is, in some sense, a limit of lengths of inscribed polygons.

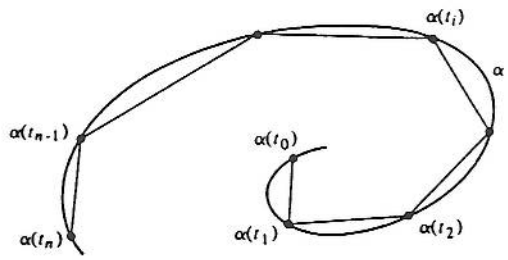

Figure 1-12

Prove that given $\epsilon > 0$ there exists $\delta > 0$ such that if $|P| < \delta$ then

$$\left| \int_a^b |\alpha'(t)|\, dt - l(\alpha, P) \right| < \epsilon.$$

Pf Recall by the definition of integral,

$$\int_a^b |\alpha'(t)| dt = \lim_{P \to 0} \Sigma (t_i - t_{i-1}) |\alpha'(t_i)| \qquad P = \max(\{t_i - t_{i-1}| \; i = 1, 2, \ldots n\})$$

i.e $\forall \varepsilon > 0$, there $\exists\, \delta' > 0$ such that if $|P| < \delta'$, then

$$\left| \left(\int_a^b |\alpha'(b)| dt \right) - \Sigma (t_i - t_{i-1}) |\alpha'(t_i)| \right| < \frac{\varepsilon}{2}.$$

On the other hand, since α' is uniformly continuous in $[a, b]$, given $\varepsilon > 0$, there $\exists\, \delta'' > 0$ such that whenever $t, s \in [a, b]$ with $|t - s| < \delta$ then $|\alpha'(t) - \alpha'(s)| < \frac{\varepsilon}{2(b-a)}$

Set $\delta = \min(\delta', \delta'')$. Then if $|P| < \delta$, we obtain,

$$\Big| \overbrace{\Sigma |\alpha(t_i) - \alpha(t_{i-1})|}^{l(\alpha, P)} - \Sigma (t_i - t_{i-1}) |\alpha'(t_i)| \Big|$$

By mean value thm $\alpha(t_i) - \alpha(t_{i-1}) = \alpha'(\xi)(t_i - t_{i-1}) \leq \sup \alpha'(s_i)$

$$\leq \Big| \Sigma (t_i - t_{i-1}) \sup_{s_i} |\alpha'(s_i)| - \Sigma (t_i - t_{i-1}) |\alpha'(t_i)| \Big|$$

$$\leq \Big| \Sigma (t_i - t_{i-1}) \sup |\alpha'(s_i) - \alpha'(t_i)| \Big| \leq (b - a) \frac{\varepsilon}{2(b-a)} = \frac{\varepsilon}{2}$$

Now $\left|\int_a^b |\alpha'(t)|\,dt - \ell(\alpha, P)\right|$

$$= \left|\int_a^b |\alpha'(t)|\,dt - \sum (t_i - t_{i-1})\,|\alpha'(t_i)| + \sum (t_i - t_{i-1})|\alpha'(t_i)| - \ell(\alpha,P)\right|$$

$$\leq \left|\int_a^b |\alpha'(t)|\,dt - \sum (t_i - t_{i-1})|\alpha'(t_i)|\right| + \left|\sum (t_i - t_{i-1})|\alpha'(t_i)| - \ell(\alpha,P)\right|$$

$$\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Q.E.D

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(10) (*Straight Lines as Shortest.*) Let $\alpha: I \longrightarrow R^3$ be a parametrized curve. Let $[a, b] \subset I$ and set $\alpha(a) = p$, $\alpha(b) = q$.

a. Show that, for any constant vector v, $|v| = 1$,

$$(q-p)\cdot v = \int_a^b \alpha'(t)\cdot v\,dt \leq \int_a^b |\alpha'(t)|\,dt.$$

b. Set

$$v = \frac{q-p}{|q-p|}$$

and show that

$$|\alpha(b) - \alpha(a)| \leq \int_a^b |\alpha'(t)|\,dt;$$

that is, the curve of shortest length from $\alpha(a)$ to $\alpha(b)$ is the straight line joining these points.

a) $$\int_a^b \alpha'(t)\cdot V \xlongequal[\alpha(t)=(x(t),\,y(t),\,z(t))]{\text{Say } V=(V_1,V_2,V_3)} \int_a^b \left(V_1 x_1'(t) + V_2\, y'(t) + V_3\, z'(t)\right) dt$$

$$= V_1 x_1(t) + V_2\, x_2(t) + V_3\, x_3(t) \Big|_a^b = V\cdot\alpha(t)\Big|_a^b = V\cdot\alpha(b) - V\cdot\alpha(a)$$

$$= V\cdot(q-p)$$

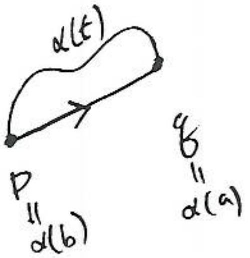
On the other hand,

$$\int_a^b \alpha'(t)\cdot v\,dt \leq \int_a^b |\alpha'(t)\cdot v|\,dt \leq \int_a^b |\alpha'(t)|\,|V|\,dt$$

$$\xlongequal{|V|=1} \int_a^b |\alpha'(t)|\,dt$$

thus we have: $(q-p)\cdot V = \int_a^b \alpha'(t)\cdot v\,dt \leq \int_a^b |\alpha'(t)|\,dt$

b) Now let $V = \dfrac{q-p}{|q-p|}$, then $|V| = 1$



By a)
$$\int_a^b |\alpha'(t)|\,dt \geq (q-p)\cdot v$$

$$= (q-p)\cdot\frac{(q-p)}{|q-p|} = \frac{|q-p|^2}{|q-p|}$$

$$= |q-p| = |\alpha(b) - \alpha(a)|.$$

Thus, the curve of shortest length from $\alpha(a)$ to $\alpha(b)$ is the straight line joining these two pts.

Sec. 1-5. The local theory of Curves Parametrized by Arc length

EXERCISES

Unless explicity stated, $\alpha: I \to R^3$ *is a curve parametrized by arc length* s, *with curvature* $k(s) \neq 0$, *for all* $s \in I$.

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1. Given the parametrized curve (helix)

$$\alpha(s) = \left(a \cos \frac{s}{c}, a \sin \frac{s}{c}, b\frac{s}{c}\right), \qquad s \in R,$$

where $c^2 = a^2 + b^2$, $a > 0,\ b > 0$

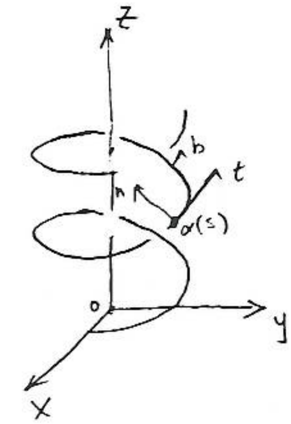
a. Show that the parameter s is the arc length.

b. Determine the curvature and the torsion of α.

c. Determine the osculating plane of α.

d. Show that the lines containing $n(s)$ and passing through $\alpha(s)$ meet the z axis under a constant angle equal to $\pi/2$.

e. Show that the tangent lines to α make a constant angle with the z axis.



a) $\alpha'(s) = \left(-\frac{a}{c}\sin\frac{s}{c},\ \frac{a}{c}\cos\frac{s}{c},\ \frac{b}{c}\right)$

$$|\alpha'(s)|^2 = \left(\frac{a}{c}\right)^2\left(\sin^2\frac{s}{c} + \cos^2\frac{s}{c}\right) + \left(\frac{b}{c}\right)^2 = \frac{a^2+b^2}{c^2} = 1$$

$\Rightarrow$ s is the arc length

b) $\alpha''(s) = \left(-\frac{a}{c^2}\cos\frac{s}{c},\ -\frac{a}{c^2}\sin\frac{s}{c},\ 0\right)$

$$|\alpha''(s)| = \sqrt{\left(\frac{a}{c^2}\right)^2} = \frac{a}{c^2} = \frac{a}{a^2+b^2}$$

$\Longrightarrow K(s) = \frac{a}{a^2+b^2} = \frac{a}{c^2},\ \forall s \in I$

$\vec{n}(s) = \frac{\alpha''(s)}{\|\alpha''(s)\|} = \left(-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0\right)$

$$\vec{b}(s) = \vec{t}(s) \times \vec{n}(s) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\frac{a}{c}\sin\frac{s}{c} & \frac{a}{c}\cos\frac{s}{c}, & \frac{b}{c} \\ -\cos\frac{s}{c} & -\sin\frac{s}{c} & 0 \end{vmatrix}$$

$$= \left[\frac{b}{c}\sin\frac{s}{c}\right]\vec{i} - \left[\frac{b}{c}\cos\frac{s}{c}\right]\vec{j} + \frac{a}{c}\vec{k}$$

$$\vec{b}'(s) = \left[\frac{b}{c^2}\cos\frac{s}{c},\ \frac{b}{c^2}\sin\frac{s}{c},\ 0\right]$$

$$= -\frac{b}{c^2}\left[-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0\right] = -\frac{b}{c^2}\vec{n}(s)$$

Therefore $\tau(s) = -\frac{b}{c^2}$

c) $n(s) = \frac{\alpha''(s)}{|\alpha''(s)|} = \frac{c^2}{a}\left(-\frac{a}{c^2}\cos\frac{s}{c},\ -\frac{a}{c^2}\sin\frac{s}{c},\ 0\right)$

$$= \left(-\cos\frac{s}{c},\ -\sin\frac{s}{c},\ 0\right)$$

$t(s) = \alpha'(s) = \left(-\frac{a}{c}\sin\frac{s}{c},\ \frac{a}{c}\cos\frac{s}{c},\ \frac{b}{c}\right)$

The osculating plane $= \mathrm{Span}\,(t(s), n(s))$

$= \{x\,t(s) + y\,n(s) \mid x, y \in \mathbb{R}\}$

d) z-axis $= \mathrm{span}\{V\}$, $V = (0,0,1)$

$$\cos\langle V, n(s)\rangle = \frac{V \cdot n}{|V||n|} = (0,0,1)\cdot\left(-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0\right) = 0$$

$\Rightarrow \langle V, n(s)\rangle = \frac{\pi}{2}$ $\qquad$ (Note: $0 \le \langle V, n(s)\rangle \le \pi$)

e) $\cos\langle V, t(s)\rangle = \dfrac{V \cdot t(s)}{|V|\,|t(s)|} = (0,0,1)\cdot\left(-\frac{a}{c}\sin\frac{s}{c},\ \frac{a}{c}\cos\frac{s}{c},\ \frac{b}{c}\right)$

$= \dfrac{b}{c}$ (where $0 \leq \langle V, t(s)\rangle \leq \pi$)

$\Rightarrow \langle V, t(s)\rangle = \arccos\dfrac{b}{c}$

<u>Note</u>: $\arccos\frac{b}{c}$ is a constant since $\cos x$ is monotone decreasing over $[0,\pi]$ and by inverse function Thm $\arccos x$ is well defined.

Thus $\langle V, t(s)\rangle$ is a constant angle, for $\forall s \in I$.

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(*2.) Show that the torsion τ of α is given by

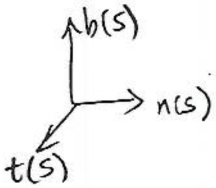
$$\tau(s) = -\frac{\alpha'(s) \wedge \alpha''(s) \cdot \alpha'''(s)}{|k(s)|^2}.$$

Pf: We are going to show $\alpha'(s) \wedge \alpha''(s) \cdot \alpha'''(s) = -\tau(s)\,\kappa^2$

Recall
$$\alpha'(s) = t(s)$$
$$\alpha''(s) = \kappa(s)\, n(s)$$
$$\alpha'''(s) = \kappa'(s)\, n(s) + \kappa(s)\, n'(s)$$
$$= \kappa'(s)\, n(s) + \kappa(s)\big(-\kappa(s)\, t(s) - \tau(s)\, b(s)\big)$$
$$= \kappa'(s)\, n(s) - \kappa^2(s)\, t(s) - \kappa(s)\tau(s)\, b(s)$$

Now $\alpha'(s) \wedge \alpha''(s) = \kappa(s)\big(t(s) \wedge n(s)\big)$

$$\big(\alpha'(s) \wedge \alpha''(s)\big) \cdot \alpha'''(s) = \kappa(s)\big(t(s) \wedge n(s)\big) \cdot \big(\kappa'(s)\, n(s) - \kappa^2(s)\, t(s) - \kappa(s)\tau(s)\, b(s)\big)$$
$$= \kappa(s)\kappa'(s)\, \overbrace{\cancel{\big(t(s) \wedge n(s)\big) \cdot n(s)}}^{0} - \kappa^3(s)\, \overbrace{\cancel{\big(t(s) \wedge n(s)\big) \cdot t(s)}}^{0} - \kappa^2(s)\,\tau(s)\,\big(t(s) \wedge n(s)\big) \cdot b(s)$$
$$= -\kappa^2(s)\,\tau(s)\, b(s) \cdot b(s) = -\kappa^2(s)\,\tau(s)\, \overbrace{|b(s)|^2}^{1} = -\kappa^2(s)\,\tau(s)$$



Thus $\tau(s) = -\dfrac{\big(\alpha'(s) \wedge \alpha''(s)\big) \cdot \alpha'''(s)}{\kappa^2(s)}$

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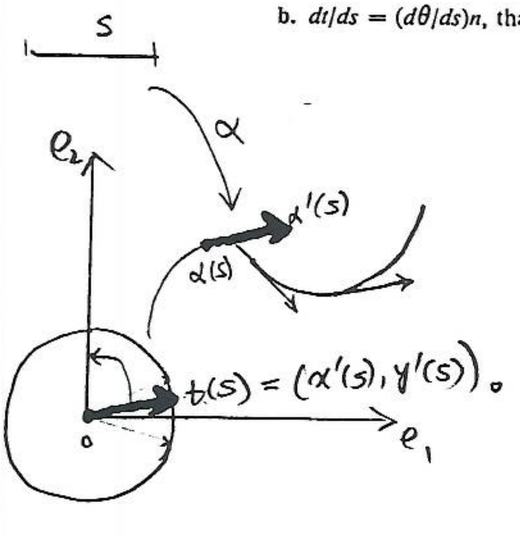
3. Assume that $\alpha(I) \subset R^2$ (i.e., α is a plane curve) and give k a sign as in the text. Transport the vectors $t(s)$ parallel to themselves in such a way that the origins of $t(s)$ agree with the origin of R^2; the end points of $t(s)$ then describe a parametrized curve $s \longrightarrow t(s)$ called the *indicatrix of tangents* of α. Let $\theta(s)$ be the angle from e_1 to $t(s)$ in the orientation of R^2. Prove (a) and (b) (notice that we are assuming that $k \neq 0$).

a. The indicatrix of tangents is a regular parametrized curve.

b. $dt/ds = (d\theta/ds)n$, that is, $k = d\theta/ds$.



a) Proof: Let $\alpha : I \to \mathbb{R}^2$ be a curve parametrized by arc length.

Then $\alpha'(s) = (x'(s), y'(s))_{\alpha(s)} = \underset{\text{at } \alpha(s)}{t(s)}$ has norm of 1.

Let's transport $t(s)$ parallel to themself in such a way that the start point of $t(s)$ agree with the origin of $\mathbb{R}^2$. We obtain a vector $t(s)$ at O.

Now since $|t(s)| = 1 \Rightarrow$ all $t(s)$ lies on a circle.

Now $t'(s) = (x''(s), y''(s))_o \Rightarrow |t'(s)| = \sqrt{(x''(s))^2 + (y''(s))^2} = K(s)$

Since we assume $K(s) \neq 0 \Rightarrow |t'(s)| \neq 0$

$\Rightarrow t(s)$ is regular

b) Now write $t(s) = (\cos\theta(s), \sin\theta(s))$

$$\Rightarrow t'(s) = (-\sin\theta(s)\,\theta'(s), \cos\theta(s)\,\theta'(s))$$

$$= (-\sin\theta(s), \cos\theta(s))\theta'(s)$$

$$= \frac{d\theta}{ds} n(s)$$

$$\Rightarrow K(s) = \frac{d\theta}{ds}.$$

Q.E.D.

page 23 (5.) A regular parametrized curve α has the property that all its tangent lines pass through a fixed point.

a. Prove that the trace of α is a (segment of a) straight line.

b. Does the conclusion in part a still hold if α is not regular?

a) Let p be the fixed pt. Let α be parametrized by arclength ($s \in I$)

A tangent line at $\alpha(s)$ is a line with direction $\alpha'(s)$ and passing a point $\alpha(s)$:

$$l(t) = \alpha'(s)t + \alpha(s).$$

By hypothesis, for each choise of s, there exists $t(s)$ (t depends on the choice of s) such that $\alpha'(s)\,t(s) + \alpha(s) = p$. $\cdots (*)$

Then $t(s)$ that satisfies this equation must be a differential function of s since $|\alpha'(s)| \neq 0$. (In fact,

$\alpha'(s)\,t(s) = p - \alpha(s) \Rightarrow \alpha'(s)\cdot\alpha'(s)\,t(s) = (p-\alpha(s))\cdot\alpha'(s)$

$\Rightarrow t(s) = \dfrac{(p-\alpha(s))\cdot\alpha'(s)}{|\alpha'(s)|^2}$.)

Let's differentiate $(*)$ both sides with respect to s,

$\Rightarrow \alpha''(s)\,t(s) + \alpha'(s)\,t'(s) + \alpha'(s) = 0$

$\Rightarrow t(s)\,\alpha''(s) + (t'(s)+1)\,\alpha'(s) = 0 \cdots (**)$

Now since α is parametrized by arclength $\Rightarrow \alpha'(s) \perp \alpha''(s)$

$\Rightarrow$ either $\alpha'(s)$ and $\alpha''(s)$ linearly-independent or $\alpha''(s) = 0$ (since $\alpha'(s) \neq 0$)

But if $\alpha'(s), \alpha''(s)$ linearly indep $\Rightarrow t(s) = 0$ and $t'(s) + 1 = 0$

That is a contradiction. Thus we must have $\alpha''(s) = 0$

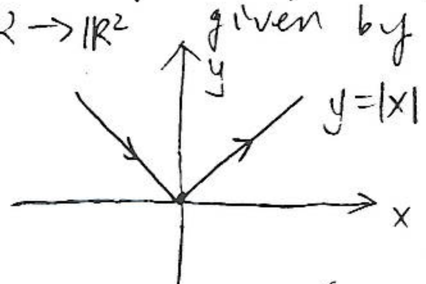
$\Rightarrow \alpha'(s) = \vec{c} \Rightarrow \alpha(s) = s\vec{c} + \vec{d}$, where $\vec{c}, \vec{d}$ are constant vector.

$\Rightarrow \alpha(s)$ is a (segment of a) straight line.

b) The conclusion does not hold if α is not regular.

How to make an example?

First note: a) fails for a non parametric curve $\alpha : \mathbb{R} \to \mathbb{R}^2$ given by $\alpha(t) = (t, |t|)$



all the tangent line of α pass through 0 (except at 0, the tangent line is not well defined).

We are going to modify α to make it to be a parametrized curve.

Let's consider

$$\beta(t) = \begin{cases} (0, 0), & t = 0 \\ (e^{\frac{1}{t}}, -e^{\frac{1}{t}}), & t < 0 \\ (e^{-\frac{1}{t}}, e^{-\frac{1}{t}}), & t > 0 \end{cases}$$

$B(t)$ on x, y plane is a curve $y = |x|$. But now

$$x(t) = \begin{cases} e^{\frac{1}{t}}, & t < 0 \\ 0, & t = 0 \\ e^{-\frac{1}{t}}, & t > 0 \end{cases}, \qquad y(t) = \begin{cases} -e^{-\frac{1}{t}} & t < 0 \\ 0 & t = 0 \\ e^{-\frac{1}{t}}, & t > 0 \end{cases}$$

both smooth. (i.e. Infinitely differentiable)

It suffices to check $x(t), y(t)$ are differentiable at 0:

For $x(t)$: $\lim_{t \to -0} (e^{\frac{1}{t}})' = \lim_{t \to -0} -\frac{1}{t^2} e^{\frac{1}{t}} = 0$, $\lim_{t \to +0} (e^{-\frac{1}{t}})' = \lim_{t \to +0} \frac{1}{t^2} e^{-\frac{1}{t}} = 0$

Similary for $y(t)$. (In fact all of its derivatives are 0 at $t = 0$ since exponential function tends to 0 much fast that a polynomial function tends zero.). $\Rightarrow \beta(t)$ is a differentiable parametrized curve which is not a line segment but with all tangent lines pass through the origin.

Q.E.D

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(6.) A *translation* by a vector v in R^3 is the map $A: R^3 \longrightarrow R^3$ that is given by $A(p) = p + v$, $p \in R^3$. A linear map $\rho: R^3 \longrightarrow R^3$ is an *orthogonal transformation* when $\rho u \cdot \rho v = u \cdot v$ for all vectors $u, v \in R^3$. A *rigid motion* in R^3 is the result of composing a translation with an orthogonal transformation with positive determinant (this last condition is included because we expect rigid motions to preserve orientation).

a. Demonstrate that the norm of a vector and the angle θ between two vectors, $0 \leq \theta \leq \pi$, are invariant under orthogonal transformations with positive determinant.

b. Show that the vector product of two vectors is invariant under orthogonal transformations with positive determinant. Is the assertion still true if we drop the condition on the determinant?

c. Show that the arc length, the curvature, and the torsion of a parametrized curve are (whenever defined) invariant under rigid motions.

a) Let $u = v$ in $\rho u \cdot \rho v = u \cdot v \Rightarrow |\rho v|^2 = |v|^2 \Rightarrow |\rho v| = |v|$

$\cos\langle \rho u, \rho v\rangle = \frac{\rho u \cdot \rho v}{|\rho u|\,|\rho v|} = \frac{u \cdot v}{|u|\,|v|} = \cos\langle u, v\rangle \Rightarrow$ angle $\langle \rho u, \rho v\rangle$ = angle $\langle u, v\rangle$ if $0 \leq \theta \leq \pi$.

b) By a proposition from our lecture, we have:

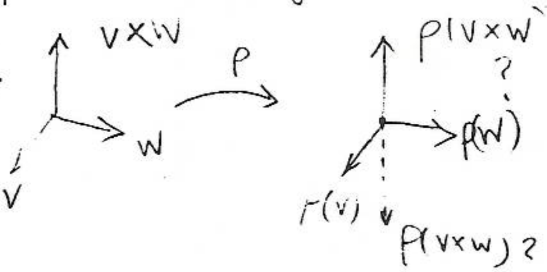
Let $v, w \in \mathbb{E}^3$. Let ρ be an orthogonal transformation of $\mathbb{E}^3$

Then $\rho(v \times w) = (\det \rho)(\rho v \times \rho w)$.

$\because \det \rho = 1 \Rightarrow \rho(v \times w) = \rho v \times \rho w$ (i.e what we want to show)

(As you may recall the proof of above proposition used algebraic method.

Let's prove b) again in a geometric method.



Since ρ is an orthonormal transformation, it preserves angles. Now since $v \times w$ perpendicular to v and $w \Rightarrow \rho(v \times w)$ perpendicular to $\rho(v)$ and $\rho(w)$. Hence $\rho(v \times w)$ perpendicular to the plane spanned by $\rho(v)$, $\rho(w)$. Now $\rho(v) \times \rho(w)$ also perpendicular to the plane spanned by $\rho(v)$ and $\rho(w)$.

Moreover $|\rho(v) \times \rho(w)| = |\rho(v)|\,|\rho(w)| \sin\langle \rho(v), \rho(w)\rangle$

$= |v|\,|w| \sin\langle v, w\rangle = |v \times w| = |\rho(v \times w)|$

Thus $\rho(v) \times \rho(w) = \pm \rho(v \times w) \cdots (*)$

Now since $\det \rho = 1 \Rightarrow \rho$ keeps orientation of a basis of $\mathbb{R}^3$

Let $x = v \times w$. If $\{v, w, x\}$ is right hand-sided $\Rightarrow \det(v, w, x) = 1$

Now $\det(\rho v, \rho w, \rho x) = \det \rho \cdot \det(v, w, x) = 1 \cdot 1 = 1 \Rightarrow \rho(v), \rho(w), \rho(x)$ also right hand.

Since $\{v, w, v \times w\}$ is a basis w/ positive orientation in $\mathbb{R}^3$ $\Rightarrow$ so is $\{\rho(v), \rho(w), \rho(v \times w)\}$

Now $\{\rho v, \rho w, \rho(v) \times \rho(w)\}$ is a basis of positive orientation.

$(*)$ implies $\rho(v \times w) = \rho(v) \times \rho(w)$.

Let A be a translation by v in $\mathbb{R}^3$

c) Then $A(\alpha(t)) = \alpha(t) + v \Rightarrow [A(\alpha(t))]'_t = (\alpha(t)+v)'_t = \alpha'(t)$

Let ρ be an O.N transformation with positive determinant

$$\rho(\alpha(t)) = \begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \rho_{11}x(t) + \rho_{12}y(t) + \rho_{13}z(t) \\ \rho_{21}x(t) + \rho_{22}y(t) + \rho_{23}z(t) \\ \rho_{31}x(t) + \rho_{32}y(t) + \rho_{33}z(t) \end{bmatrix}$$

$$[\rho(\alpha(t))]'_t = \begin{bmatrix} \rho_{11}x'(t) + \rho_{12}y'(t) + \rho_{13}z'(t) \\ \rho_{21}x'(t) + \rho_{22}y'(t) + \rho_{23}z'(t) \\ \rho_{31}x'(t) + \rho_{32}y'(t) + \rho_{33}z'(t) \end{bmatrix} = \begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{bmatrix} \begin{bmatrix} x'(t) \\ y'(t) \\ z'(t) \end{bmatrix}$$

$$= \rho\alpha'(t)$$

Further, we have $[A(\alpha(t))]''_t = A\alpha''(t)$, $[A(\alpha(t))]''' = A\alpha'''(t)$

$$[\rho\alpha(t)]''_t = \rho\alpha''(t), \quad [\rho\alpha(t)]'''_t = \rho\alpha'''(t)$$

Therefore arclength is preserved under the rigid motion R since

$$\int_{t_0}^{t} |(R\alpha(s))'|\, ds = \int_{t_0}^{t} |R\alpha'(s)|\, ds = \int_{t_0}^{t} |\alpha'(s)|\, ds$$

Note: A rigid motion also preserves the regularity of a curve.

$$|(A\alpha(s))'| = |A\alpha'(s)| = |\alpha'(s)| \neq 0$$

$$\Rightarrow (A\alpha(s))' \neq 0$$

Now the curvature of α is preserved since

$$|(A\alpha(s))''| = |A\alpha''(s)| = |\alpha''(s)| \quad (\text{i.e } K_{A\alpha(s)} = K_{\alpha(s)})$$

and torsion is preserved since

$$\tau_{A(\alpha(s))} = -\frac{[(A\alpha(s))' \times (A\alpha(s))''] \cdot (A\alpha(s))'''}{K_{A\alpha(s)}} = -\frac{[A\alpha'(s) \times A\alpha''(s)] \cdot A\alpha'''}{K_{A\alpha(s)}}$$

$$= -\frac{A(\alpha'(s) \times \alpha''(s)) \cdot A\alpha'''(s)}{K_{A\alpha(s)}} = -\frac{[\alpha'(s) \times \alpha''(s)] \cdot \alpha'''(s)}{K_{\alpha(s)}} = \tau_0$$

Page 23 (*7) Let $\alpha: I \longrightarrow R^2$ be a regular parametrized plane curve (arbitrary parameter), and define $n = n(t)$ and $k = k(t)$ as in Remark 1. Assume that $k(t) \neq 0$, $t \in I$. In this situation, the curve

$$\beta(t) = \alpha(t) + \frac{1}{k(t)} n(t), \qquad t \in I,$$

is called the *evolute* of α (Fig. 1-17).

a. Show that the tangent at t of the evolute of α is the normal to α at t.

b. Consider the normal lines of α at two neighboring points $t_1, t_2, t_1 \neq t_2$. Let t_1 approach t_2 and show that the intersection points of the normals converge to a point on the trace of the evolute of α.

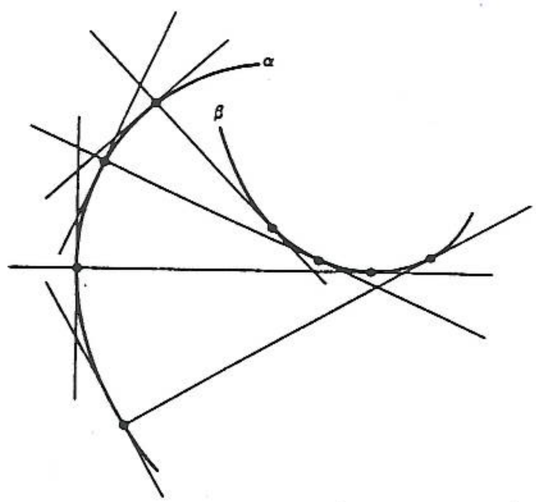


Figure 1-17

a) Without of loss of generality, we can assume α is paramtrized by arc length. [Since we always can reparametrize α by arclength: $\bar{\alpha}(s) = \alpha(t(s))$. Then we can write the evolute of $\alpha(t(s))$

$$\beta(t(s)) = \alpha(t(s)) + \frac{1}{K(t(s))} n(t(s))$$

as $\bar{\beta}(s) = \bar{\alpha}(s) + \frac{1}{\bar{k}(s)} \bar{n}(s)$.]

Now let's differentiate the evolute of $\alpha(s)$:

$$\beta(s) = \alpha(s) + \frac{1}{K(s)} n(s) \qquad s \in I$$

both sides with respect to s.

$$\Rightarrow \beta'(s) = \alpha'(s) + \frac{-K(s)}{K^2(s)} n(s) + \frac{1}{K(s)} n'(s) \quad \cdots \circledast$$

We wish to show $\alpha'(s) \perp \beta'(s)$.

Let's dot $\circledast$ both sides by $\alpha'(s)$

$\because n(s) \perp t(s)$ $(t(s) = \alpha'(s))$ 0

C plane

$$\Rightarrow \beta'(s) \cdot \alpha'(s) = \alpha'(s) \cdot \alpha'(s) + \frac{-K(s)}{K^2(s)} \cancel{n(s) \cdot \alpha'(s)} + \frac{1}{\cancel{K(s)}} (-\cancel{K(s)} t(s) - \cancel{\tau(s) b(s)}) \cdot \alpha'$$

$$= |t(s)|^2 - |t(s)|^2 = 0$$

b) Parametrize α by arc length. The normalizes at s_1 and are

$$L_1(\lambda) = \alpha(s_1) + \lambda n(s_1) \quad , \quad L_2(\sigma) = \alpha(s_2) + \sigma n(s_2) \, , \quad \lambda, \sigma$$

To solve the intersection, we let $L_1(\lambda) = L_2(\sigma)$. This gives u

$$\alpha(s_1) + \lambda n(s_1) = \alpha(s_2) + \sigma n(s_2)$$

$$\Rightarrow \alpha(s_2) - \alpha(s_1) = \lambda n(s_1) - \sigma n(s_2)$$

divide by $s_2 - s_1$

$$\Rightarrow \frac{\alpha(s_2)-\alpha(s_1)}{s_2-s_1} = \frac{\lambda n(s_1) - \sigma n(s_2)}{s_2 - s_1}$$

$$\Rightarrow \frac{\alpha(s_2)-\alpha(s_1)}{s_2-s_1} = \frac{\lambda n(s_1) - \sigma n(s_2) + \sigma n(s_1) - \sigma n(s_1)}{s_2-s_1}$$

$$\Rightarrow \frac{\alpha(s_2)-\alpha(s_1)}{s_2-s_1} = -\frac{\sigma(n(s_2)-n(s_1))}{s_2-s_1} + \frac{(\lambda-\sigma)n(s_1)}{s_2-s_1}$$

Let's dot both sides by $\alpha'(s_1)$.

$\because n(s_1) \perp \alpha'(s_1)$

$$\Rightarrow \left(\frac{\alpha(s_2)-\alpha(s_1)}{s_2-s_1}\right) \cdot \alpha'(s_1) = -\sigma \frac{n(s_2)-n(s_1)}{s_2-s_1} \cdot \alpha'(s_1) + \frac{(\lambda-\sigma)\cancel{n(s_1)\cdot\alpha'(s_1)}}{s_2-s_1}$$

Now Let's take the limit both sides for $s_2 \to s_1$.

$$\Rightarrow \lim_{s_2\to s_1} \frac{\alpha(s_2)-\alpha(s_1)}{s_2-s_1} \cdot \alpha'(s_1) = -\left(\lim_{s_2\to s_1}\sigma\right) \lim_{s_2\to s_1} \frac{n(s_2)-n(s_1)}{s_2-s_1} \cdot \alpha'(s_1)$$

$$\Rightarrow \alpha'(s_1)\cdot\alpha'(s_1) = -\left(\lim_{s_2\to s_1}\sigma\right) n'(s_1)\cdot\alpha'(s_1)$$

$\because$ plane curve

$$\Rightarrow |\alpha'(s_1)|^2 = -\left(\lim_{s_2\to s_1}\sigma\right)\left(-K(s_1)t(s_1) - \underbrace{\cancel{\tau(s_1)}}_{0} b(s_1)\right)\alpha'(s_1)$$

$$\Rightarrow |\alpha'(s_1)|^2 = \left(\lim_{s_2\to s_1}\sigma\right) K(s_1)\cdot|\alpha'(s_1)|^2$$

$|\alpha'(s_1)| = 1$

$$\Rightarrow \lim_{s_2\to s_1}\sigma = \frac{1}{K}$$

Thus the intersection pt is $L_2(\sigma) = \alpha(s_2) + \frac{1}{K} n(s_2)$ which is

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8. The trace of the parametrized curve (arbitrary parameter)

$$\alpha(t) = (t, \cosh t), \qquad t \in R,$$

is called the *catenary*.

a. Show that the signed curvature (cf. Remark 1) of the catenary is

$$k(t) = \frac{1}{\cosh^2 t}.$$

b. Show that the evolute (cf. Exercise 7) of the catenary is

$$\beta(t) = (t - \sinh t \cosh t, 2 \cosh t).$$

a) To show $K(t) = \frac{1}{\cosh^2 t}$, we can use prob. 12 (Last HW assignm

$$K(t) = \frac{|\alpha'(t) \times \alpha''(t)|}{|\alpha'(t)|^3}$$

Now $\alpha'(t) = (1, \sinh t) \Rightarrow |\alpha'(t)|^2 = \sqrt{1+\sinh^2 t} = \cosh t$

$\alpha''(t) = (0, \cosh t)$

$$\Rightarrow \alpha' \times \alpha'' = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & \sinh t & 0 \\ 0 & \cosh t & 0 \end{vmatrix} = (\cosh t)\vec{k} \Rightarrow |\alpha' \times \alpha''| = \cosh t$$

Thus $K(t) = \frac{\cosh t}{\cosh^3 t} = \frac{1}{\cosh^2 t}$

Another way to show a) is by directly reparametrizing $\alpha(t)$ by arc length.

Let $s(t) = \int_0^t |\alpha'(t)|\,dt = \int_0^t \sqrt{1^2 + (\sinh t)^2}\,dt = \int_0^t \cosh t\,dt = \sinh t$

$\Rightarrow S(t) = \sinh t \Rightarrow t\ \sinh^{-1}(s)$ ∴ Let's denote this inverse function by $t = g(s)$. $\left(\begin{matrix} g(S(t)) = t \\ \& \ g'(S(t)) = \frac{1}{S'(t)} \end{matrix}\right)$ Let $\beta = \alpha \circ g$. Then $|\beta'(s)| = 1$

$\beta(s) = (g(s), \cosh(g(s)))$

$\beta'(s) = (g'(s), \sinh(g(s))\, g'(s)) = (g'(s), s g'(s))$.

$\beta''(s) = (g''(s), g'(s) + s g''(s))$. There, for α

$K(t) n(t) = (g''(s(t)), g'(s(t)) + S(t) g''(s(t)))$

(We want to compute $\kappa(t)$ for $\alpha(t)$. So we want to consid[er] β'' at the pt $\alpha(t)$. But this is $\beta(s(t))$ since
$$\beta(s(t)) = \alpha(g(s(t))) = \alpha(t) \quad (\text{Recall } g = s^{-1}).$$

Notice that $g(s(t)) = t \Rightarrow g'(s(t)) = \dfrac{1}{s'(t)} = \dfrac{1}{\cosh t}$

and $g''(s(t_0)) = \dfrac{-s''(t)}{(s'(t))^3} = \dfrac{-s(t)}{\cosh^3(t)}$

$$\boxed{\begin{aligned} &g(s(t)) = t \\ &\Rightarrow \frac{dg}{ds}\cdot\frac{ds}{dt} = 1 \Rightarrow \frac{dg}{ds} = \frac{1}{\frac{ds}{dt}} \\ &\Rightarrow \frac{d^2g}{ds^2}\cdot\frac{ds}{dt}\cdot\frac{ds}{dt} + \frac{dg}{ds}\frac{d^2s}{dt^2} = 0 \\ &\Rightarrow \frac{d^2g}{ds^2} = -\frac{dg}{ds}\frac{d^2s}{dt^2}\Big/\left(\frac{ds}{dt}\right)^2 = -\frac{d^2 s/dt^2}{(ds/dt)^3} \end{aligned}}$$

$$\Rightarrow \beta''(s(t)) = \left(\frac{-s(t)}{\cosh^3 t},\ \frac{1}{\cosh t} - \frac{(s(t))^2}{\cosh^3(t)}\right)$$
$$= \left(\frac{-\sinh t}{\cosh^3(t)}\ \ \frac{\cosh^2 t - \sinh^2 t}{\cosh^3(t)}\right)$$
$$= \left(\frac{-\sinh t}{\cosh^3(t)},\ \frac{1}{\cosh^3(t)}\right)$$

$$\Rightarrow |\beta''(s(t))| = \frac{1}{\cosh^2 t}$$

$$\Rightarrow \pm n(t) = \frac{\beta''(s(t))}{|\beta''(s(t))|} = \left(\frac{-\sinh t}{\cosh^3 t}\cdot\cosh^2 t,\ \frac{1}{\cosh^3(t)}\cosh^2 t\right)$$
$$= \left(-\frac{\sinh t}{\cosh t},\ \frac{1}{\cosh t}\right) \triangleq V$$

But $\det\left(\beta'(s(t))\ \ V\right) = \begin{vmatrix} \frac{1}{\cosh t} & -\frac{\sinh t}{\cosh t} \\ \frac{\sinh t}{\cosh t} & \frac{1}{\cosh t} \end{vmatrix} = \dfrac{1}{\cosh^2 t} + \dfrac{\sinh^2 t}{\cosh^2 t} = 1$

$$\boxed{\begin{aligned} \beta'(s(t)) &= (g'(s(t)),\ s(t)g'(s(t))) \\ &= \left(\frac{1}{s'(t)},\ \frac{s(t)}{s'(t)}\right) \\ &= \left(\frac{1}{\cosh t},\ \frac{\sinh t}{\cosh t}\right) \end{aligned}}$$

Thus $n(t) = \left(-\dfrac{\sinh t}{\cosh t},\ \dfrac{1}{\cosh t}\right)$

Now $\beta''(s(t)) = \dfrac{1}{\cosh^2 t}\, n(t)$

$\Rightarrow \kappa(t) = \dfrac{1}{\cosh^2 t}$ is the signed curvatur[e].

b) The evolute of the catenary is
$$\beta(t) = \alpha(t) + \frac{1}{\kappa(t)}n(t) = (t, \cosh t) + \cosh^2 t\left(-\frac{\sinh t}{\cosh t},\ \frac{1}{\cosh t}\right)$$
$$= (t - \sinh t\cosh t,\ 2\cosh t).$$

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(12) Let $\alpha: I \longrightarrow R^3$ be a regular parametrized curve (not necessarily by arc length) and let $\beta: J \longrightarrow R^3$ be a reparametrization of $\alpha(I)$ by the arc length $s = s(t)$, measured from $t_0 \in I$ (see Remark 2). Let $t = t(s)$ be the inverse function of s and set $d\alpha/dt = \alpha'$, $d^2\alpha/dt^2 = \alpha''$, etc. Prove that

a. $dt/ds = 1/|\alpha'|$, $d^2t/ds^2 = -(\alpha'\cdot\alpha''/|\alpha'|^4)$.

b. The curvature of α at $t \in I$ is

$$k(t) = \frac{|\alpha' \wedge \alpha''|}{|\alpha'|^3}.$$

c. The torsion of α at $t \in I$ is

$$\tau(t) = -\frac{(\alpha' \wedge \alpha'')\cdot\alpha'''}{|\alpha' \wedge \alpha''|^2}.$$

d. If $\alpha: I \longrightarrow R^2$ is a plane curve $\alpha(t) = (x(t), y(t))$, the signed curvature (see Remark 1) of α at t is

$$k(t) = \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}}.$$

a) Recall $s(t) = \int_{t_0}^{t} |\alpha'(t)|\,dt$

$$\frac{ds}{dt} = |\alpha'(t)| \Rightarrow \frac{dt}{ds} = \frac{1}{|\alpha'(t)|} = \frac{1}{\alpha'} \qquad \alpha' \triangleq \alpha'(t)$$

$$\frac{d^2t}{ds^2} = \frac{d}{ds}\left(\frac{1}{|\alpha'(t(s))|}\right) = -\frac{\frac{d}{ds}|\alpha'(t(s))|}{|\alpha'(t(s))|^2}$$

$$\frac{d}{ds}\left(\sqrt{(x'(t(s)))^2+(y'(t(s)))^2+(z'(t(s)))^2}\right) = \frac{1}{2}\frac{2x'(t(s))x''(t(s))\frac{dt}{ds} + 2(\quad) + 2(}{\sqrt{x'^2+y'^2+z'^2}}$$

$$= \frac{\alpha'\cdot\alpha''\frac{dt}{ds}}{|\alpha'|} = \frac{\alpha'\cdot\alpha''}{|\alpha'|^2}$$

$$\Rightarrow \frac{d^2t}{ds^2} = -\frac{\alpha'\cdot\alpha''}{|\alpha'|^4}$$

b) $K_\alpha(t) = K_\beta(s) = |\beta''(s)|$

$$\beta = \alpha(t(s)) \Rightarrow \frac{d\beta}{ds} = \frac{d\alpha}{dt}\frac{dt}{ds}$$

$$\frac{d^2\beta}{ds^2} = \frac{d^2\alpha}{dt^2}\left(\frac{dt}{ds}\right)^2 + \frac{d\alpha}{dt}\frac{d^2t}{ds^2}$$

$$= \alpha''(t)\frac{1}{|\alpha'(t)|^2} - \frac{\alpha'(t)(\alpha'\cdot\alpha'')}{|\alpha'(t)|^4}$$

i.e. $$\beta''(s) = \frac{(\alpha'\cdot\alpha')\alpha'' - (\alpha'\cdot\alpha'')\alpha'}{|\alpha'(t)|^4} = \frac{(\alpha'\times\alpha'')\times\alpha'}{|\alpha'|^4}$$

$$|\beta''(s)| = \frac{|\alpha'\times\alpha''|\,|\alpha'|\sin\langle\alpha'\times\alpha'', \alpha'\rangle}{|\alpha'|^4} = \frac{|\alpha'\times\alpha''|}{|\alpha'|^3}, \text{ Since}$$

$$\sin\langle\alpha'\times\alpha'', \alpha'\rangle = \sin\frac{\pi}{2} = 1$$

c) Let $\beta(s) = \alpha(t(s))$ be a repremetrize $\alpha(t)$ by arclength. We may use Ex 2, p. 22 to compute the torsion. We apply the formula to $\beta(s)$. We wish to compute $-\dfrac{(\beta'(s) \wedge \beta''(s)) \cdot \beta'''(s)}{|K(s)|^2}$

$$\beta' = \frac{\alpha'}{|\alpha'|} \quad \text{and} \quad \beta'' = \frac{\alpha''(\alpha'\cdot\alpha') - \alpha'(\alpha'\cdot\alpha'')}{|\alpha'|^4}, \text{ so}$$

$$\beta' \wedge \beta'' = \frac{\alpha'}{|\alpha'|} \wedge \frac{\alpha''(\alpha'\cdot\alpha')}{|\alpha'|^4} = \frac{\alpha'\cdot\alpha'(\alpha'\wedge\alpha'')}{|\alpha'|^5} = \frac{1}{|\alpha'|^3}(\alpha'\wedge\alpha'')$$

We next compute $\beta'''(s)$

$$\beta'''(s) = \frac{d^3\alpha}{dt^3}\frac{dt}{ds}\left(\frac{dt}{ds}\right)^2 + \frac{d^2\alpha}{dt^2}\,2\left(\frac{dt}{ds}\right)\frac{d^2t}{ds^2} + \frac{d\alpha}{dt}\frac{dt}{ds}\frac{d^2t}{ds^2} + \frac{d\alpha}{dt}\frac{d^3t}{ds^3}$$

$$= \alpha'''\frac{1}{|\alpha'|^3} + 2\alpha''\frac{1}{|\alpha'|}\left(-\frac{\alpha'\cdot\alpha''}{|\alpha'|^4}\right) + \alpha'\frac{1}{|\alpha'|}\left(-\frac{\alpha'\cdot\alpha''}{|\alpha'|^4}\right) + \alpha'\frac{d^3t}{ds^3}$$

Note: All of these terms, except $\alpha'''\frac{1}{|\alpha'|^3}$ are zero when dotted with $\beta'(s)\times\beta''(s)$ (Since $(u\wedge v)\cdot u = (u\wedge w)\cdot v = 0$)

we do not need calculate this term

Thus $(\beta'(s)\wedge\beta''(s))\cdot\beta'''(s) = \frac{1}{|\alpha'|^3}(\alpha'\wedge\alpha'')\cdot\alpha'''\frac{1}{|\alpha'|^3}$

$$\tau_\beta(s) = -\frac{\beta'(s)\wedge\beta''(s)\cdot\beta'''(s)}{|K(s)|^2} = -\frac{1}{\cancel{|\alpha'|^3}}(\alpha'\wedge\alpha'')\cdot\alpha'''\frac{1}{\cancel{|\alpha'|^3}}\cdot\frac{(\cancel{|\alpha'|^3})^2}{|\alpha'\wedge\alpha''|^2}$$

$\parallel$

$\tau_\alpha(t)$

$$= -\frac{(\alpha'\wedge\alpha'')\cdot\alpha'''}{|\alpha'\wedge\alpha''|^2}$$

d) Apply the formula in (b) to α, where $\alpha(t) = (x(t), y(t), 0)$

Then $$k = \frac{|\alpha' \wedge \alpha''|}{|\alpha'|^3} = \frac{|x''y'' - y'x''|}{((x')^2+(y')^2)^{3/2}}.$$

The signed curvature is $\pm$ of this value. Since we require that $\{t, n\}$ be a positively oriented basis the determinent $\begin{vmatrix} x' & y' \\ \frac{x''}{k_s} & \frac{y''}{k_s} \end{vmatrix} > 0 \Rightarrow \frac{x'y'' - y''x'}{k_s} > 0$

$\Rightarrow x'y'' - y''x'$ and k_s have the same sign

$$\Rightarrow k_s = \frac{x'y'' - y''x'}{((x')^2+(y')^2)^{3/2}}$$

Q.E.D